

KEVIN LU

PhD Student, Khoury College of Computer Sciences, Northeastern University
lu.kev@northeastern.edu · kevinlu4588.github.io · [Google Scholar](https://scholar.google.com/citations?user=kevinlu4588) · github.com/kevinlu4588

RESEARCH INTERESTS

Mechanistic interpretability of biological foundation models (protein structure prediction, protein language models, and cell models), and using what these models have learned to accelerate drug discovery.

EDUCATION

Northeastern University 2026 – present
Ph.D. in Computer Science Boston, MA

Advisors: David Bau and Wengong Jin · NSF Graduate Research Fellow

Northeastern University 2022 – 2026
B.S. in Computer Science and Mathematics, Honors Program Boston, MA
GPA: 3.96/4.00

PUBLICATIONS

- [1] **Kevin Lu**, Jannik Brinkmann, Stefan Huber, Aaron Mueller, Yonatan Belinkov, David Bau, Chris Wendler. *Two Stages of Folding: Convergent Mechanisms in AI Protein Folding Trunks*. Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
- [2] **Kevin Lu**, Nicky Kriplani, Rohit Gandikota, Minh Pham, David Bau, Chinmay Hegde, Niv Cohen. *When Are Concepts Erased From Diffusion Models?* Conference on Neural Information Processing Systems (**NeurIPS**), 2025.

EXPERIENCE

OpenAI Apr 2026 – May 2026
Red Teaming Contractor Remote

- Analyzed the behavior of LLM agents for AI safety evaluation.

Takeda Pharmaceuticals Jun 2025 – Dec 2025
Machine Learning Research Intern Cambridge, MA

- Trained sparse autoencoders on ESM-2 to extract interpretable antibody features and used feature steering to generate 10,000+ candidates; 192 variants were experimentally validated, with a 20% increase in binding specificity.
- Fine-tuned ESM-2 on 1.2M antibody sequences, improving CDR reconstruction accuracy by 32%.

Babel Street Jul 2024 – Dec 2024
Machine Learning Engineer Co-op Boston, MA

- Built a multithreaded pipeline that filters, embeds, and indexes 1 TB/week of multilingual news into an Elasticsearch vector database.
- Developed a multi-agent LLM annotation system (LangGraph) with human-in-the-loop feedback, cutting annotation costs by 25%.

Neural Systems Group, Harvard Medical School Jun 2023 – May 2024
Undergraduate Researcher Boston, MA

- Built Kalman-filter features from ECG waveforms and trained SVR models for blood pressure prediction, reaching AAMI Grade B accuracy.

FELLOWSHIPS AND AWARDS

NSF Graduate Research Fellowship	3-year fellowship, \$159,000 total support	2026
Honors Global Support Fund	\$6,000 research grant	2025
Khoury Undergraduate Travel Award (2×)	\$1,000 conference travel grant	2025
John Martinson Honors Scholarship	\$40,000/year merit scholarship	2022 – 2026

TALKS AND PRESENTATIONS

Flatiron Institute , New York, NY	<i>Mechanisms of AI Protein Folding</i>	Feb 2026
New England Mechanistic Interpretability Workshop (NEMI)	<i>To Bind or Not to Bind</i>	Aug 2025
CVPR Workshop on Generative Models for Vision	<i>Where Do Erased Concepts Go?</i>	Jun 2025

SERVICE

Conference Reviewer: NeurIPS 2026, ICLR 2026
Workshop Reviewer: ICLR 2026 (Scientific Methods for DL), NeurIPS 2025 (Mechanistic Interpretability)

LEADERSHIP AND OUTREACH

President , Northeastern Veritas Forum	2023 – 2026
Community Lead , InterVarsity Christian Fellowship	2023 – 2026
Editor, Web Developer, and Science Writer , NU Sci Magazine	2022 – 2024
Teaching Aide , Dr. Martin Luther King Jr. K–8 School	2023
Elder Service Intern , Action for Boston Community Development	2022 – 2023